MAGNUS AI

AI-Powered Skill Intelligence & Learning Platform

VIDEO PRODUCTION MANUAL & USER GUIDE

Smart India Hackathon 2026 | Problem Statement PS 26101
Ministry of Statistics & Programme Implementation (MoSPI)
Data Informatics & Innovation Division (DIID) | Theme: Smart Education

VIDEO STRUCTURED IN THREE PARTS:

Part 1 Learner Role	Part 2 Trainer Role	Part 3 Admin Role

How to Use This Manual

This document is your complete script and reference guide for recording the Magnus AI demonstration video. It covers every screen, every action, and every narration point you need to confidently walk through the application across all three user roles. Read each section **before** opening your screen recorder, then follow the steps in sequence during filming.

Part	Role	Estimated Duration	Key Screens Covered
1	Learner (Ananya Sharma, JSO)	4-5 minutes	Login, Profile, Dashboard, Gap Analysis, Learning Path, Assessment, AI Practice Quiz
2	Trainer (Dr. Rahul Verma)	2-3 minutes	Login, Upload PDF Material, AI MCQ Generation, Review & Approve Quiz
3	Admin (Vikram Singh)	2-3 minutes	Login, Workforce Analytics, User Role Management, System Health & Audit Logs

Recording Tip	Use OBS Studio or Loom at 1920x1080. Narrate clearly. Keep mouse movements slow. Pause 3 seconds on each key screen. Do one full dry-run before recording.

Technology Stack

Briefly explain the tech stack in the first 45 seconds of your video. This impresses judges with engineering depth.

Frontend

Framework	React 19 + TypeScript — type-safe, component-based UI architecture.
Build Tool	Vite — ultra-fast bundler with hot module replacement.
Styling	Tailwind CSS — utility-first design tokens. Deep navy + saffron red + forest green brand palette.
State	React Context API (AuthProvider) + useState/useEffect local component state.
Routing	React Router v7 — declarative protected routes with role-based navigation guards.
Charting	Recharts — Radar charts (skill gap), Bar charts (admin analytics), responsive containers.
Icons	Lucide React — consistent SVG icon library (Target, BookOpen, Brain, Shield, etc.)
Auth & DB	Supabase — PostgreSQL, Supabase Auth (JWT), Row Level Security (RLS) enforced at DB level.

Backend (FastAPI + Python)

Framework	FastAPI — async Python REST API, auto OpenAPI docs, CORS configured.
AI / LLM	Google Gemini 3.5 Flash Lite via langchain-google-genai. Structured output (Pydantic).
RAG	LangChain orchestrates: Retrieve context from pgvector -> Augment prompt -> Generate MCQs.
Vector Store	Supabase pgvector — stores document chunk embeddings for cosine similarity retrieval.
PDF Parsing	PyMuPDF (fitz) — extracts and chunks text from uploaded PDF documents.

Validation	Pydantic v2 — type-safe request/response schemas and structured LLM output.
Server	Uvicorn ASGI server. Procfile-ready for cloud deployment.

Key API Endpoints

Endpoint	Method	Description
GET /health	GET	Server liveness check. Returns {status: ok}.
POST /api/upload	POST	Accepts PDF, parses with PyMuPDF, stores embeddings in pgvector. Returns total_chunks.
POST /api/generate-questions	POST	Takes competency_name, difficulty_level, num_questions. RAG retrieval + Gemini call -> structured MCQs.

Architecture Flow

Browser -> **React+Vite** -> **FastAPI** -> **Gemini LLM** -> **pgvector** -> **Supabase DB**

User browser -> React/Vite Frontend -> FastAPI Backend (Python) -> Google Gemini API + Supabase PostgreSQL + pgvector (Vector DB)

Demo Persona

Introduce this persona before logging in at the start of Part 1.

Field	Value
Full Name	Ananya Sharma
Designation	Junior Statistical Officer (JSO)
Department	Labour Statistics — MoSPI
Current Assignment	Annual Survey of Industries (ASI)
Experience	2 years
Education	M.Sc. Statistics, University of Delhi
Previous Training	Official Statistics Foundation Course (NSSTA), 2023
Identified Gaps	Sampling: L2 (need L3) | Python: L1 (need L2) | SQL: L2 (need L3)
Demo Email	ananya.s@mospi.gov.in

Narration	"Meet Ananya Sharma, a Junior Statistical Officer at MoSPI's Labour Statistics division. She has 2 years of experience, but her role requires Working-level proficiency in Sampling, Python, and SQL — areas where she currently has verified skill gaps. Magnus AI is here to identify those gaps precisely, give her a personalized learning path, and track her progress as she closes each gap."

L PART 1 — Learner Role

The complete closed learning loop from a JSO official's perspective.

The Learner demonstrates the **core value proposition** of Magnus AI. This is the most important part of your video.

Login -> Dashboard -> Profile -> Competency -> Learning Path -> Assessment -> AI Quiz -> Result

Step 1 — Login & Authentication

Route: /login | **Component:** Login.tsx, AuthProvider.tsx

1 Open the App

Navigate to the Magnus AI URL. Show the professional government-style login screen with Magnus AI branding.

2 Enter Credentials

Type Ananya's email (ananya.s@mospi.gov.in) and password. Click Sign In. Auth is handled by Supabase JWT tokens.

3 Role-Based Redirect

After authentication, the AuthProvider reads the user role. Learner is redirected to /learner. Attempting to visit /admin shows an Unauthorized error page — this is RBAC in action.

Step 2 — Learner Dashboard

Route: /learner | **Component:** LearnerDashboard.tsx, GapAnalysis.tsx

Walk through each dashboard element:

- **Welcome Header:** 'Welcome back, Ananya.' with 'Official Learner | JSO Track' and 'On Track' badge. Name comes from the authenticated session.
- **3 KPI Cards:** Current Level Avg (L2.8, target L3.0) | Learning Hours (42h, +4h this week) | Courses Completed (7, 3 active certificates).
- **Competency Gap Radar Chart:** Dual-radar (GapAnalysis component) — orange = Required Level, green = Current Level. Visual gap is immediately obvious for Sampling, Python, SQL.

- **Priority Growth Areas list:** Three gap cards below the chart. Each shows: Competency name, Required: L3, Current: L2, Gap: -1 Level badge.
- **AI Practice Quiz CTA:** 'Configure a Custom Practice Quiz' card at the bottom with 'Generate Quiz' button linking to /learner/quizzes.

Narration for Dashboard	"As soon as Ananya logs in, she sees her complete competency picture. The radar chart is built by comparing her assessment results against the role_competencies matrix for a Junior Statistical Officer. No guesswork — it's data-driven and explainable. She immediately knows she needs to focus on Sampling, Python, and SQL."

Step 3 — Competency Passport & Profile

Route: /learner/profile | **Components:** LearnerProfile.tsx, ProfileForm.tsx, CompetencyPassport.tsx

- **Competency Passport card:** Visual identity — Designation, Department, Job Role, Years of Experience, Education, Previous Training (NSSTA Foundation Course), Certifications.
- **Profile Edit Form:** Fields — Full Name, Employee Code, Designation, Job Role (dropdown: JSO / SO / Data Analyst / SSO / Deputy Director), Department (dropdown: Labour Statistics / National Accounts / Agricultural / Price Statistics / Methodology / IT Operations), Current Assignment, Years of Experience, Education Summary, Previous Training Programmes (textarea), Certifications.
- **Changing Job Role** recalculates which competency matrix applies — the gap analysis updates automatically.
- **Save Profile Changes:** 700ms loading spinner, then green success banner: 'Your Competency Passport has been updated.' Discard button resets to last saved state.

Step 4 — Full Competency Matrix

Route: /learner/competency | **Component:** CompetencyMatrix.tsx

Shows the complete MoSPI competency framework mapped to Ananya's role:

- **4 Domains:** Statistical (Survey Design, Sampling, National Accounts...), Technical (Python, R, SQL, GIS, AI/ML...), Digital Governance (Cybersecurity, Data Privacy...), Behavioural & Managerial (Leadership, Ethics...).
- **5 Levels:** L1 Awareness, L2 Basic, L3 Working, L4 Advanced, L5 Expert.
- Each competency shows current level, required level, and gap. Colour-coded: green = meets requirement, amber = partial gap, red = critical gap.

Step 5 — Learning Pathways & Resource Catalogue

Route: /learner/learning-path | **Components:** LearningPathPage.tsx, ResourceCard.tsx

- **AI Recommendations tab (default):** Banner — '3 Gaps Targeted — tailored for Sampling, Python, SQL.' Resources from iGOT Karmayogi, NSSTA, and TPAC ranked by relevance to Ananya's specific gaps.
- **Resource Catalog tab:** Full searchable catalogue. Live search by keyword — demonstrate typing 'Python' to filter.
- **Resource Card:** Title, Source (iGOT / NSSTA / TPAC badge), Competency addressed, Difficulty (L1-L5), Duration, Match % (why it was recommended), CTA button.

Narration for Learning Path	"Magnus AI doesn't show a generic course catalogue. The recommendation engine looks at Ananya's specific gaps at her current level and ranks resources from iGOT Karmayogi, NSSTA, and TPAC that directly close those deficiencies. Every recommendation card explains exactly why it was selected — this is explainable AI in practice."

Step 6 — Official Competency Assessment

Routes: /learner/assessments + /learner/assessment/:id | **Components:** AssessmentPage.tsx, AssessmentRunner.tsx

Open an assessment (e.g., 'Sampling Theory Level 2 Evaluation'):

- **Assessment Runner:** Progress bar (Question 1 of 10), countdown timer, clear question text, 4 radio-button options. Keyboard accessible (Tab to navigate, Space to select).
- **No answers shown before submission** — the AssessmentRunner deliberately withholds correct answers until the quiz is submitted.
- **Result Screen:** Trophy icon, animated score bar (0% -> final %, 1-second CSS transition), Final Score card (e.g., 75% in green), Awarded Level card (e.g., L3 in brand red).
- **Score colour coding:** Green >= 75% Merit, Amber >= 50% Pass, Red < 50% Needs Improvement.
- **'Competency updated in Passport'** — the Awarded Level is written back to user_competencies in Supabase.
- **Next Steps:** 'View Updated Passport' button -> /learner/profile. 'Return to Dashboard' button -> /learner.

Step 7 — AI-Powered Practice Quiz (KEY DIFFERENTIATOR)

Route: /learner/quizzes | **Component:** LearnerQuizzes.tsx

Highlight this prominently — it demonstrates the live Gemini RAG pipeline.

Show the Quiz Configurator form:

- **Target Competency dropdown:** Two groups — 'Recommended for You (Identified Gaps)' shows Sampling, Python, SQL with star prefix. 'All Competencies' lists the full framework. Select Python.
- **Toughness Level:** L1 Awareness through L5 Expert. Select L2 Basic.

- **Number of Questions:** 5 / 10 / 15 / 20. Select 5.
- **Auto Time Allocation:** Info box auto-calculates: 5 questions x 1.5 min = 7-minute time limit. Auto-submit on expiry.
- **Click 'Generate & Start Quiz':** Button shows spinning loader: 'Generating Questions...'

Explain the backend pipeline during the loading state:

1 RAG Query

Frontend calls POST /api/generate-questions. FastAPI queries pgvector with: 'Python concepts for a L2 Basic proficiency assessment'. Top-6 relevant document chunks retrieved by cosine similarity.

2 Prompt Construction

LangChain builds a system prompt injecting: retrieved context chunks, competency name (Python), difficulty (L2), and number of questions (5). Instructs Gemini to base questions ONLY on this context.

3 Gemini Structured Output

ChatGoogleGenerativeAI.with_structured_output(QuestionGenerationResult) forces the LLM to return a perfectly typed Pydantic model. No JSON parsing errors. Each question has: question_text, 4 MCQOption objects (text + is_correct bool), explanation, difficulty, competency.

4 Quiz Rendered

Frontend maps the Pydantic response to AssessmentRunner's question format. Quiz begins with a timer. Learner answers, submits, sees score.

Offline Fallback	If the FastAPI backend is unreachable, the frontend catches the error and loads placeholder [OFFLINE MODE] questions. The demo always works. Notification: 'Backend unreachable — Start the FastAPI server for real Gemini RAG generation.'

PART 2 — Trainer Role

Upload PDF -> AI generates MCQs -> Trainer reviews & approves -> Published to learners.

Log out. Log in as **Dr. Rahul Verma** (r.verma@nssta.gov.in). Notice the sidebar changes — Trainers see Materials, Question Generator, Assessments, Learners, Analytics.

Login as Trainer -> Trainer Dashboard -> Upload PDF -> AI Generates MCQs -> Review Questions -> Edit / Approve

Step 1 — Trainer Dashboard

Route: /trainer | **Component:** TrainerDashboard.tsx

- Overview metrics: Materials Uploaded, Quizzes Generated, Assessments Published.
- Upload Material panel on the left, AI Quiz Verification panel on the right — both visible simultaneously.
- Different sidebar navigation from Learner role — same app, completely different experience.

Step 2 — Upload Learning Material

Route: /trainer/upload | **Component:** TrainerUploadPage.tsx, UploadMaterial.tsx

Walk through the upload panel:

- **File Dropzone:** Drag-and-drop zone. Accepts PDF, PPTX, DOCX (max 20 MB). Border turns orange on drag-over. On drop, shows filename + file size preview with an X button to clear.
- **Resource Title:** Text input. Example: 'Introduction to Survey Sampling Theory'.
- **Competency Domain:** Dropdown — Statistical, Technical, Digital Governance, Behavioural. Selecting a domain filters the Target Competency dropdown dynamically.
- **Target Competency:** Filtered by selected domain. Example: Statistical domain -> Survey Design, Sampling, National Accounts, etc.
- **Target Difficulty Level:** L1 Awareness through L5 Expert.

Click 'Parse & Generate Quizzes ->':

1. **Upload & Parse**

   File bytes sent to POST /api/upload. PyMuPDF reads the PDF, splits into chunks. Each chunk stored as a vector embedding in Supabase pgvector. Returns total_chunks count.

2 Question Generation

Immediately calls POST /api/generate-questions with competency name, difficulty, document filename, and num_questions=3.

3 Structured Output

Gemini returns QuestionGenerationResult Pydantic model. Each GeneratedQuestion has question_text, list of 4 MCQOption (text + is_correct bool), explanation, difficulty, competency.

4 Success Notification

Green banner: 'Success! Processed X chunks and generated Y AI MCQs.' A PendingQuiz is added to the Review panel with status='pending'.

Offline Fallback	If the backend is offline, one pre-written domain-relevant question is generated as a fallback. Notification: 'Material processed. (Live Gemini RAG available when backend is connected).' Demo always works.

Step 3 — AI Quiz Verification: Review & Approve

Route: /trainer/review | **Component:** TrainerReviewPage.tsx, ReviewQuizzes.tsx

The review interface is a **two-panel layout**:

- **Left panel — Quiz List:** All generated quizzes. Amber badge for PENDING, green for APPROVED. Badge shows 'X Pending Reviews'. Click a quiz to open it.
- **Right panel — Question Detail:** Quiz title, source material filename, question cards.
- **Question card (read mode):** Question text, 4 options (correct option highlighted green with filled circle), Official Explanation in a beige box.
- **Edit mode (hover pencil icon):** Textarea for question text, radio buttons to select correct answer, text inputs to edit option text, textarea for explanation. 'Save Changes' / 'Cancel' buttons.
- **'Approve All Questions' button:** Green button at top of detail panel. Only visible for PENDING quizzes. Clicking changes status to APPROVED. Badge count decrements.

Narration for Trainer	"The Trainer never writes a single question manually. They upload their NSSTA syllabus PDF and Gemini reads it through our RAG pipeline, extracts key concepts at the right difficulty level, and generates MCQs grounded in that specific material. The Trainer's job is quality control — review, edit if needed, approve. This is how Magnus AI scales content creation while maintaining pedagogical accuracy."

A PART 3 — Administrator Role

Workforce intelligence, user role management, system health & audit compliance.

Log out. Log in as **Vikram Singh** (v.singh@mospi.gov.in, role: admin). Admin layout has different sidebar: Overview, Workforce, Skill Gaps, Training, Analytics, Settings.

Login as Admin	->	**Admin Dashboard**	->	**Workforce Analytics**	->	**User Roles**	->	**System Health**	->	**Audit Logs**

Step 1 — Admin Dashboard: Top KPI Cards

Route: /admin | **Component:** AdminDashboard.tsx

- **Total Enrolled: 4,281** — Active learners across all MoSPI departments.
- **Active Trainers: 142** — Trainers managing content on the platform.
- **Critical Skill Gaps: 847** — Org-wide count of competencies where workforce level is >= 2 below required.
- **System Uptime: 99.9%** — Platform reliability metric.
- Three segmented tabs below: Workforce Analytics | User Roles | System Health.

Step 2 — Workforce Analytics (Charts)

Component: AdminDashboard.tsx analytics tab — Recharts library

- **Organization Skill Gaps — Radar Chart (left):** Dual radar. Grey polygon = Required Level. Orange/red polygon = Actual Workforce Level. Subjects: Python, SQL, Sampling, Data Viz, Machine Learning. Hover tooltips show exact values. The gap between polygons = training priority. Python and Sampling show largest gaps.
- **Training & Assessment Throughput — Bar Chart (right):** Grouped bars for Jan–Jun. Orange bars = Courses Completed. Green bars = Assessments Passed. Shows organizational learning velocity over time. Hover tooltips and legend.

Narration for Analytics	"The Admin sees aggregated, anonymized workforce intelligence — no individual PII. This radar chart tells the MoSPI Capacity Building division exactly where to direct NSSTA training programmes next quarter. Python and Sampling are the two biggest org-wide gaps. That's a strategic budget decision backed by real data."

Step 3 — User Role Management

Route: /admin/users | **Component:** UserManagement.tsx

- **Search bar:** Real-time filter by name, email, or department. Type 'Labour' — only Labour Statistics users appear.
- **Filter dropdown:** 'All Roles' | Learners | Trainers | Administrators. Select 'Trainers' — active filter chip appears: 'Filtered by: trainer — 2 users found'.
- **User table:** Avatar (initials), Full Name, Email (@mospi.gov.in / @nssta.gov.in), Department, Role badge (dark = Admin, orange = Trainer, green = Learner), Last Active timestamp.
- **Demo: Change Role:** Click 'Change Role' on Priya Patel (Learner). Role badge becomes a dropdown. Select 'Trainer'. Click Done. Badge instantly changes to orange Trainer badge.
- In production, this writes to the profiles table in Supabase. RLS policies enforce the new permissions on the very next request.

Step 4 — System Health & Audit Logs

Route: /admin/health | **Component:** SystemHealth.tsx

Telemetry cards (top row):

- API Gateway — ONLINE (animated green pulse dot), 45ms avg, 0.01% error rate.
- Vector Database — SYNCED, 24,592 document chunks indexed in pgvector.
- Backend Compute — 34% LOAD, 4 Cores Active, horizontal progress bar.
- Storage Bucket — 12% USED, 1.2 GB / 10 GB, progress bar.

System Audit Log table:

- Immutable event trail: assessments completed, RAG pipeline syncs, role changes, document uploads, LLM timeouts, database backups.
- Columns: Timestamp, Actor (person or System), Action, Entity Type, Status badge (SUCCESS / WARNING / ERROR).
- **Filter buttons:** All Events | Warnings | Errors. Click 'Errors' — only the 'LLM Timeout / Gemini API' error event appears with a red badge.
- Sticky header — stays visible while scrolling through logs.

Narration for System Health	"For a government platform, the Audit Log is not optional — it's a compliance requirement. Every significant action is timestamped and attributed: role changes, document uploads, assessment attempts, LLM requests. This gives administrators full accountability and traceability across the entire platform."

Video Script Outline

Use this as your spoken script guide. Total target length: 8-12 minutes.

Time	Section	What You Say / Do
0:00	Intro & Problem Statement	Introduce yourself and team. State the SIH problem: officials in India's OSS lack a tool to identify their exact skill gaps and get personalized guidance. Introduce Magnus AI.
0:45	Tech Stack (30 sec)	Show the stack: React+Vite frontend, FastAPI backend, Google Gemini for AI, Supabase PostgreSQL + pgvector for RAG. Show architecture flow.
1:15	Demo Persona Introduction	Introduce Ananya Sharma — JSO at MoSPI Labour Statistics, 2 years exp, gaps in Sampling, Python, SQL.
1:35	Login as Learner	Open Magnus AI. Log in as Ananya. Show role-based redirect to Learner Dashboard.
2:10	Learner Dashboard	Walk through KPI cards. Explain the dual-radar gap chart. Highlight the 3 gap cards.
3:00	Profile / Competency Passport	Open Profile. Show Competency Passport card. Scroll through form fields. Demonstrate save.
3:45	Learning Pathways	Navigate to Learning Path. Show AI Recommendations tab. Explain why each resource was selected. Switch to Catalog, demonstrate search.
4:30	Official Assessment	Open a pre-seeded assessment. Show question interface + timer. Submit. Show animated result screen with score and Awarded Level.
5:15	AI Practice Quiz (KEY)	Navigate to Practice Quizzes. Configure: Python, L2, 5 questions. Click Generate. Show loading spinner. Take the quiz. Show result.
6:30	Login as Trainer	Log out. Log in as Dr. Rahul Verma. Show different sidebar.
6:45	Upload Learning Material	Upload a statistics PDF. Fill metadata form. Click Parse & Generate. Show spinner then success notification.
7:30	Review & Approve Quiz	Navigate to Review. Select pending quiz. Show questions. Edit one question. Approve all. Badge count changes to 0.
8:15	Login as Admin	Log out. Log in as Vikram Singh. Show admin KPI dashboard.
8:30	Workforce Analytics	Show radar chart (org skill gaps). Show bar chart (training throughput). Explain strategic insights.
9:15	User Role Management	Search users. Filter by role. Demonstrate role change on a user. Mention RLS enforcement.

9:50	System Health & Audit Logs	Show telemetry cards. Filter audit logs to Errors. Explain compliance importance.
10:30	Closing — The Learning Loop	Summarize the closed loop: Profile -> Assessment -> Gap -> Recommendation -> Learning -> Quiz -> Competency Update -> New Recommendation. Thank viewers.

Key Points for Judges

Work these into your narration naturally. They address the SIH evaluation criteria.

[ ] Competency-First, Not Course-First

Magnus AI starts with WHO the learner is (role, current level, department) before recommending WHAT to learn. Most LMS platforms show a catalogue. We show a competency gap first, then specific resources to close it.

[ ] Explainability in Every Recommendation

Every recommendation card tells the learner WHY it was recommended — which specific gap it addresses, what level it targets, and the match percentage. No black-box suggestions. Judges can see the reasoning.

[ ] Government-Interoperability Ready

The integration adapter pattern means iGOT Karmayogi and NSSTA/TPAC data is consumed through a standardized interface. When official API credentials become available, only the adapter needs to change — zero changes to business logic.

[ ] RAG-Powered, Grounded AI (No Hallucination)

Questions are grounded in the actual uploaded documents. Gemini generates MCQs based ONLY on the source material. This prevents hallucination and ensures domain-relevant, accurate assessments.

[ ] Security & Privacy by Design

RLS policies in Supabase ensure learners cannot access each other's data. Admins see aggregated metrics, not individual PII. No API keys or secrets are sent to the frontend. All server-side operations are authenticated.

[ ] Free-Tier Stack, Production-Ready Architecture

Built entirely on free-tier services: Supabase free tier, Gemini API free quota, Vercel free tier. The modular architecture supports scaling to government production infrastructure without architectural changes.

[ ] Closed Learning Loop

Profile -> Assessment -> Gap -> Recommendation -> Learning -> Quiz -> Competency Update -> New Recommendation. Magnus AI is continuous. Every completed quiz updates the Competency Passport and refreshes recommendations automatically.

Quick Reference — Routes & Components

Role	Route	Component	What's There
All	/login	Login.tsx	Email/password login. Supabase Auth. Role-based redirect.
Learner	/learner	LearnerDashboard.tsx	KPI cards, Skill Gap Radar, Practice Quiz CTA.
Learner	/learner/profile	LearnerProfile.tsx	Competency Passport + Profile Edit Form (10 fields).
Learner	/learner/competency	LearnerCompetency.tsx	Full 4-domain, 5-level competency matrix.
Learner	/learner/learning-path	LearningPathPage.tsx	AI Recommendations tab + Searchable Resource Catalog.
Learner	/learner/assessment/:id	AssessmentPage.tsx	Take assessment, animated result, competency update.
Learner	/learner/quizzes	LearnerQuizzes.tsx	Configure + Generate AI Practice Quiz via Gemini RAG.
Trainer	/trainer	TrainerDashboard.tsx	Overview metrics + Upload + Review panels.
Trainer	/trainer/upload	TrainerUploadPage.tsx	Upload PDF, configure competency/domain/difficulty, generate.
Trainer	/trainer/review	TrainerReviewPage.tsx	Two-panel: quiz list (left) + question editor (right).
Admin	/admin	AdminDashboard.tsx	KPI cards + Workforce Analytics charts.
Admin	/admin/users	AdminDashboard.tsx	User Role Management: search, filter, change roles.
Admin	/admin/health	AdminDashboard.tsx	Telemetry cards + Audit Log with status filters.

Pre-Recording Checklist

- [ ] Frontend running: `cd frontend && npm run dev` (http://localhost:5173)
- [ ] Backend running: `cd backend && uvicorn app.main:app --reload` (http://localhost:8000)
- [ ] Test /health: Visit http://localhost:8000/health -> should return {'status': 'ok'}
- [ ] GEMINI_API_KEY is set in backend/.env

- [ ] All three demo accounts seeded in Supabase (Ananya = learner, Dr. Rahul = trainer, Vikram = admin)
- [ ] Demo PDF ready for upload (any statistics-related document, e.g., NSSTA syllabus)
- [ ] Screen recorder at 1920x1080, audio clear, mic tested
- [ ] Browser in full-screen, extensions disabled (no notification pop-ups)
- [ ] Full dry-run done end-to-end before final recording

Magnus AI — Competency-first. Explainable. Role-aware. Interoperable. Learning-loop driven.

Smart India Hackathon 2026 | PS 26101 | Ministry of Statistics & Programme Implementation